Software Engineering Lab-1

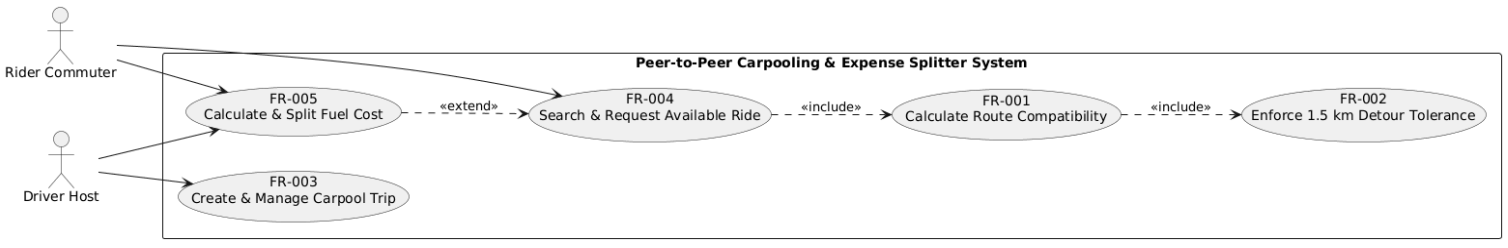
Name: Ayaan Ahmed
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Section: B

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall calculate route compatibility between a driver and rider based on their commute paths.	High	Pass if compatible routes are identified; Fail if incompatible routes are accepted.	Prevents unsuitable carpool matches.
FR-002	Functional	The system shall enforce a maximum 1.5 km detour tolerance when matching a rider with a driver.	High	Pass: Only matches with a detour of 1.5 km or less are accepted. Fail: A match exceeding 1.5 km is accepted.	Ensures that drivers do not have to make unreasonable detours to pick up riders.
FR-003	Functional	The system shall allow drivers to create and manage carpool trips by specifying their route	High	Pass: A driver can create, modify, and remove a trip with valid route and availability details. Fail: A trip with missing or invalid required information is created.	Allows drivers to advertise available rides and keep their trip information up to date.
FR-004	Functional	The system shall allow riders to search for and request available rides based on their pickup location	High	Pass: The system displays compatible available rides and allows the rider to submit a request. Fail: The system displays rides that do not satisfy the specified route or travel requirements.	Helps riders find suitable carpooling options that match their commute needs.

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FR-005	Functional	The system shall automatically calculate and split the fuel cost among the driver and participating riders for a confirmed carpool trip.	High	Pass: The system calculates the total fuel cost and displays each participant's share. Fail: The system produces an incorrect or missing expense split.	Provides a fair and transparent method for sharing travel expenses.
NFR-001	Non Functional	User ratings, driving license verification documents, and travel histories must be stored securely with encrypted backups.	High	Pass: Security testing confirms that sensitive data is encrypted during storage and backup, and unauthorized users cannot access it. Fail: Sensitive information is stored without encryption or can be accessed by unauthorized users.	Protects sensitive user information and prevents unauthorized access or data loss.
NFR-002	Non Functional	The system shall provide route matching and expense calculation results within 3 seconds under normal operating conditions and simulated peak load.	High	Pass: At least 95% of route matching and expense calculation requests are completed within 3 seconds during performance testing. Fail: More than 5% of requests exceed the 3-second response time.	Ensures that users receive matches and expense information quickly enough for a smooth experience.



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Use-Case Flow Specification

Use Case: Search & Request Available Ride

Use Case ID: UC-004
Related Functional Requirement: FR-004
Primary Actor: Rider Commuter
Supporting Actor: Driver Host

Preconditions

1. The rider is registered and logged into the system.
2. The rider has entered their pickup location, destination, preferred date, and travel time.
3. At least one driver has created an available carpool trip in the system.

Postconditions

1. A suitable carpool ride is identified based on the rider's route and travel requirements.
2. The rider's ride request is submitted to the selected driver.
3. The system records the ride request and displays its status to the rider.
4. The applicable fuel cost share can be calculated for the requested ride.

Main Success Scenario

1. The rider enters their pickup location, destination, preferred date, and travel time.
2. The system searches for available driver trips matching the rider's travel requirements.
3. The system compares the rider's route with each available driver's route.
4. The system calculates the additional detour required for the driver to accommodate the rider.
5. The system checks that the calculated detour does not exceed the maximum **1.5 km detour tolerance**.
6. The system displays the compatible rides to the rider.
7. The rider selects a suitable ride.
8. The rider submits a ride request to the driver.
9. The system records the request and notifies the driver.
10. The system displays the request status to the rider.

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11. Once the ride is confirmed, the system calculates the rider's applicable share of the fuel cost.

Alternate Flow – Detour Exceeds Tolerance

A1. At Step 5, if the calculated driver detour exceeds **1.5 km**, the system rejects that driver as an incompatible match.

A2. The system continues searching for other available drivers whose routes satisfy the detour tolerance.

A3. If another compatible driver is found, the system displays the ride as an available option.

A4. If no compatible driver is found, the system informs the rider that no suitable carpool ride is currently available.